ProjFlow: Projection Sampling with Flow Matching for Zero-Shot Exact Spatial Motion Control

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Abstract

Precise spatial control of human motion generation—such as following joint trajectories, matching keyframes, or lifting 2D sketches into 3D—typically requires task-specific retraining or slow inference-time optimization that still fails to satisfy constraints exactly. We present ***ProjFlow****, a training-free projection sampler for flow-matching human motion models that achieves* exact *constraint satisfaction without any fine-tuning or inner-loop optimization. The key idea is to cast spatial motion control as a linear inverse problem and enforce constraints by projecting each flow step's clean motion estimate onto the feasible set using a kinematics-aware metric. This metric, derived from a Laplacian-based skeletal topology, spreads corrections coherently along the skeleton, preserving motion naturalness while guaranteeing hard constraint satisfaction. For sparse or long-gap inpainting, we introduce time-varying pseudo-observations with dynamic masking and adaptive variance, enabling robust guidance without retraining. ProjFlow is evaluated on HumanML3D across trajectory control, all-joint control, and 2D-to-3D motion lifting. It achieves* exact *zero trajectory, location, and average errors (0.0000) across all spatial control tasks while delivering competitive FID scores (0.107 for pelvis, 0.097 for all joints), surpassing zero-shot baselines and matching the realism of task-specific trained methods. On 2D-to-3D lifting, ProjFlow attains zero 2D reprojection error and outperforms Sketch2Anim [28] by a large margin in FID (0.349 vs. 0.525).*

1. Introduction

Human motion generation has made remarkable strides with the advent of diffusion models [5, 20, 26] and, more recently, flow matching models [3, 17]. Yet enabling precise *spatial control*—constraining specific joints to follow prescribed trajectories, matching keyframe poses, or lifting 2D sketch inputs into 3D—remains a significant challenge. Current approaches fall into two broad paradigms, each with fundamental limitations.

Trained controllers. Methods such as OmniControl [23], MaskControl [18], and ACMDM+CtrlNet [17] learn dedicated control modules alongside (or on top of) generative models. While these methods yield high-fidelity outputs, they require retraining for each new type of control signal and are tied to specific control modalities. Extending them to novel constraint types—such as 2D projection constraints or composite objectives—necessitates repeated data collection and training.

Zero-shot optimizers. Diffusion-based methods like DNO [11] and GMD [10] operate at inference time without retraining by optimizing over the latent diffusion noise. Similarly, DPS [4] and DDNM [21] propose posterior sampling strategies for diffusion inverse problems. However, optimization-based zero-shot methods are computationally expensive, require many function evaluations, and often produce only approximate constraint satisfaction—leaving residual spatial errors that can degrade downstream use cases.

Exact constraint satisfaction is particularly critical in practical motion authoring pipelines, where residual positioning errors—even on the order of millimeters—can propagate through physics simulations, break object-contact interactions, or require costly post-hoc correction passes. Moreover, many downstream consumers of generated motion (renderers, game engines, inverse-kinematics solvers) assume hard skeletal constraints and may produce visible artifacts or numerical instability when these are only approximately met. The gap between approximate and exact satisfaction thus represents not merely a benchmark metric discrepancy but a genuine barrier to deploying generative motion models in interactive creative applications.

We propose **ProjFlow**, a novel training-free approach that achieves *exact* hard constraint satisfaction in flow-matching human motion models. The central insight is that many practical spatial control tasks—trajectory following, keyframe matching, upper-body editing, and 2D-to-3D lifting—can be cast as *linear inverse problems* of the form $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{\epsilon}$, admitting closed-form projections onto the feasible set.

At each step of the flow sampler, ProjFlow (i) predicts

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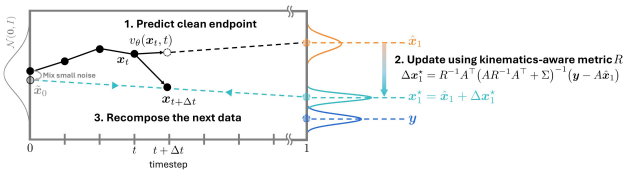

Figure 1. **Overview of the Projection Sampling Step.** At each flow step t, ProjFlow predicts the clean motion estimate $\hat{\boldsymbol{x}}_0$, applies a closed-form $\boldsymbol{R}$-norm projection to satisfy the linear constraint $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$, and recomposes the result with stochastic noise to return to the flow manifold.

the clean motion endpoint via the model's vector field, (ii) applies a closed-form correction that maps the predicted motion onto the constraint manifold using a kinematics-aware $\boldsymbol{R}$-norm, and (iii) recomposes the corrected endpoint with noise to stay on the flow manifold. The kinematics-aware metric encodes skeletal topology through a graph Laplacian, ensuring that constraint corrections propagate coherently along the skeleton rather than causing isolated joint artifacts. For sparse or long-gap inpainting, we introduce time-varying pseudo-observations: a dynamic masking scheme with adaptive variance that gradually reveals the constraint signal as sampling progresses, enabling robust guidance even for challenging inpainting tasks.

Our main contributions are:

- A unified view of spatial motion control as a linear inverse problem solvable by a closed-form projection-based flow sampler, without retraining or inner-loop optimization.
- A kinematics-aware metric derived from the skeletal graph Laplacian that distributes constraint corrections coherently across the skeleton, improving realism while preserving exact feasibility.
- Time-varying pseudo-observations with dynamic masking and adaptive variance for robust zero-shot inpainting of motion sequences with sparse or long-gap constraints.
- Extensive evaluation on HumanML3D [8] demonstrating exact constraint satisfaction (0.0000 trajectory, location, and average errors) with competitive motion quality across trajectory control, all-joint control, upper-body editing, and 2D-to-3D lifting tasks.

2. Related Work

Text-to-motion generation. Modern text-to-motion systems learn to generate diverse and realistic 3D human motions from text descriptions. MDM [20] introduced classifier-free diffusion for motion, while subsequent works improved quality and speed [5, 6, 26]. Flow matching models [2, 14, 15] have emerged as an attractive alternative, offering straighter trajectories and faster sampling; ACMDM [17] applies this paradigm to human motion and achieves strong performance on HumanML3D [8].

Trained spatial controllers. Several works train dedicated spatial control modules for motion generation. OmniControl [23] integrates joint trajectory signals via cross-attention adapters trained jointly with the motion model. MaskControl [18] proposes a masking-based control scheme that achieves exact joint constraints for a fixed set of control modes. MotionLCM [6] adapts ControlNet [25] to the motion domain for fast inference. PriorMDM [19] uses inter-diffusion priors for body-part composition. While these methods achieve high fidelity, they require retraining and are specialized to their respective control modalities.

Zero-shot and optimization-based control. GMD [10] guides a pre-trained diffusion model via classifier guidance from trajectory constraints. DNO [11] optimizes the initial noise of a pre-trained model to minimize a motion constraint objective, enabling flexible control without retraining but at high computational cost and with only approximate satisfaction. Diffusion Posterior Sampling (DPS) [4] approximates the likelihood gradient for general inverse problems in diffusion models. DDNM [21] projects within the null space of a linear operator during diffusion reverse steps, achieving exact linear constraints in image generation. FlowDPS [12] extends posterior sampling to flow matching models. ProjFlow is most closely related to DDNM and DiracDiffusion [7] in its use of closed-form projections, but adapts these ideas to the flow matching framework with a kinematics-aware metric tailored to skeletal motion.

2D-to-3D motion lifting. Sketch2Anim [28] proposes a full pipeline to transfer 2D sketch storyboards to 3D animations, including a specialized trained network. PhysDiff [24] lifts physics constraints into diffusion-based motion generation. Our 2D-to-3D lifting formulation casts orthographic projection as a linear operator and applies ProjFlow directly, without any specialized training for this task.

Motion inpainting and editing. Several works address motion inpainting and editing using diffusion models, including PriorMDM [19], TEDi [27], and SkelBetweener [1]. InterControl [22] and InterGen [13] handle multi-person interaction constraints. Our pseudo-observation strategy offers a principled alternative to masking-based inpainting within the flow matching framework.

3. Preliminaries

Human motion representation. Following HumanML3D [8], we represent human motion as a sequence of 3D joint positions $\boldsymbol{x} \in \mathbb{R}^{3\times T\times J}$, where T is the number

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of frames and $J = 22$ is the number of body joints. In the normalized model space used by ACMDM [17], joint positions are centered and scaled by dataset statistics (μ, σ).

Flow matching for motion generation. Flow matching [2, 14, 15] defines a continuous normalizing flow between a source distribution (Gaussian noise $\boldsymbol{x}_1 \sim \mathcal{N}(0, \mathbf{I})$) and a target distribution (human motions $\boldsymbol{x}_0 \sim p_{\text{data}}$). A neural network $v_\theta(\boldsymbol{x}_t, t, \boldsymbol{c})$ is trained to predict the vector field at time $t \in [0, 1]$ conditioned on text $\boldsymbol{c}$, typically via the regression objective:

$$\mathcal{L} = \mathbb{E}_{t,\boldsymbol{x}_0,\boldsymbol{x}_1}\left[\|v_\theta(\boldsymbol{x}_t, t, \boldsymbol{c}) - (\boldsymbol{x}_0 - \boldsymbol{x}_1)\|^2\right], \quad (1)$$

where $\boldsymbol{x}_t = (1-t)\boldsymbol{x}_1 + t\boldsymbol{x}_0$ is the linear interpolant. At sampling time, an ODE solver integrates v_θ from $t = 1$ to $t = 0$ to obtain a clean sample. Given the vector field prediction $v_\theta(\boldsymbol{x}_t, t)$, the predicted clean endpoint at time t is:

$$\hat{\boldsymbol{x}}_0(t) = \boldsymbol{x}_t + (1-t) \cdot v_\theta(\boldsymbol{x}_t, t, \boldsymbol{c}). \quad (2)$$

Linear inverse problems. A linear inverse problem is defined by a measurement model:

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{\epsilon}, \quad (3)$$

where $\boldsymbol{A} \in \mathbb{R}^{m \times n}$ is the forward operator, $\boldsymbol{y} \in \mathbb{R}^m$ are the measurements, and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma_\epsilon^2 \mathbf{I})$ is optional Gaussian noise. In the noiseless case ($\sigma_\epsilon \to 0$), we seek a solution satisfying $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{y}$ exactly. The minimum $\boldsymbol{R}$-norm solution, for a positive definite metric $\boldsymbol{R}$, is given by:

$$\hat{\boldsymbol{x}} = \boldsymbol{x} + \boldsymbol{R}^{-1}\boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{R}^{-1}\boldsymbol{A}^\top)^{-1}(\boldsymbol{y} - \boldsymbol{A}\boldsymbol{x}). \quad (4)$$

This is the orthogonal projection of $\boldsymbol{x}$ onto the affine constraint set $\{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{y}\}$ in the $\boldsymbol{R}$-weighted inner product.

4. Method: ProjFlow

ProjFlow integrates constraint satisfaction directly into the flow matching sampling loop through a closed-form projection step. Figure 1 provides an overview.

4.1. Spatial Control as a Linear Inverse Problem

Many practical motion control tasks can be expressed as linear constraints on the joint positions $\boldsymbol{x}$:

Trajectory and keyframe control. Requiring joint j to follow a trajectory $\{\boldsymbol{y}_t^j\}_{t=1}^T$ corresponds to selecting specific (t, j, axis) entries of $\boldsymbol{x}$. The forward operator $\boldsymbol{A}$ is a binary selection (masking) matrix.

Upper-body editing. Locking the upper-body joints to a reference sequence is a special case of joint-wise masking.

2D-to-3D lifting. Given 2D projections $\boldsymbol{y}_t^j \in \mathbb{R}^2$ of joint j from an orthographic camera $\boldsymbol{P} \in \mathbb{R}^{2\times 3}$, the constraint is $\boldsymbol{y}_t^j = \boldsymbol{P}\boldsymbol{x}_t^j$. Stacking across frames and joints yields a linear system of the form equation 3.

In all cases, the measurements $\boldsymbol{y}$ can be expressed as $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$, where $\boldsymbol{A}$ encodes the appropriate selection and/or projection operations. The sampler operates in the normalized model space, so $\boldsymbol{A}$ and $\boldsymbol{y}$ are converted accordingly (see section 4.6).

4.2. Kinematics-Aware Projection

A critical design choice is the metric $\boldsymbol{R}$ used in the projection equation 4. A naive Euclidean metric ($\boldsymbol{R} = \mathbf{I}$) corrects only the constrained joints and degrees of freedom, leaving the unconstrained joints unchanged. This leads to abrupt discontinuities at constrained joints and unnatural motion artifacts, as nearby skeleton joints are not adjusted to accommodate the correction.

We propose a **kinematics-aware metric** $\boldsymbol{R}$ that encodes the skeletal topology of the human body. Concretely, we define $\boldsymbol{R}$ using the graph Laplacian $\boldsymbol{L}$ of the skeleton graph:

$$\boldsymbol{R} = (\boldsymbol{L} \otimes \mathbf{I}_3)^\alpha + \lambda \mathbf{I}, \quad (5)$$

where $\boldsymbol{L} \in \mathbb{R}^{J\times J}$ is the normalized Laplacian of the skeleton connectivity graph, $\otimes$ denotes the Kronecker product (acting over joints and spatial dimensions), $\alpha > 0$ controls the smoothness strength, and $\lambda > 0$ is a regularization constant for invertibility. By penalizing differences between neighboring joints according to the skeleton graph, the metric ensures that constraint corrections propagate *coherently along bones and kinematic chains*, producing corrections that resemble natural skeletal deformations.

The projection step for a predicted clean motion $\hat{\boldsymbol{x}}_0$ becomes:

$$\hat{\boldsymbol{x}}_0^* = \hat{\boldsymbol{x}}_0 + \boldsymbol{R}^{-1}\boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{R}^{-1}\boldsymbol{A}^\top + \sigma_\epsilon^2\mathbf{I})^{-1}(\boldsymbol{y} - \boldsymbol{A}\hat{\boldsymbol{x}}_0). \quad (6)$$

For exact constraints ($\sigma_\epsilon = 0$), this reduces to the exact projection; for noisy measurements, it implements the minimum mean squared error (MMSE) estimate. In practice, the normal matrix $(\boldsymbol{A}\boldsymbol{R}^{-1}\boldsymbol{A}^\top + \sigma_\epsilon^2\mathbf{I})^{-1}$ is precomputed once via Cholesky decomposition for efficiency, as the operator $\boldsymbol{A}$ is fixed during sampling.

4.3. Stochastic Recomposition

After projecting the clean estimate $\hat{\boldsymbol{x}}_0^*$, a naïve approach would set $\boldsymbol{x}_{t-\Delta t} = \hat{\boldsymbol{x}}_0^*$ and proceed. However, this collapses all stochasticity at the first projection step, leading to overdetermined trajectories and degenerate samples with high FID.

Instead, ProjFlow uses a **stochastic recomposition** step that mixes the projected clean estimate with fresh noise to return to the flow manifold at time $t - \Delta t$:

$$\boldsymbol{x}_{t-\Delta t} = (1 - (t - \Delta t))\,\boldsymbol{x}_1' + (t - \Delta t)\,\hat{\boldsymbol{x}}_0^*, \quad (7)$$

where $\boldsymbol{x}_1' \sim \mathcal{N}(0, \mathbf{I})$ is resampled at each step and mixed according to the flow schedule. The amount of noise injected is controlled by a schedule η_t that decays towards

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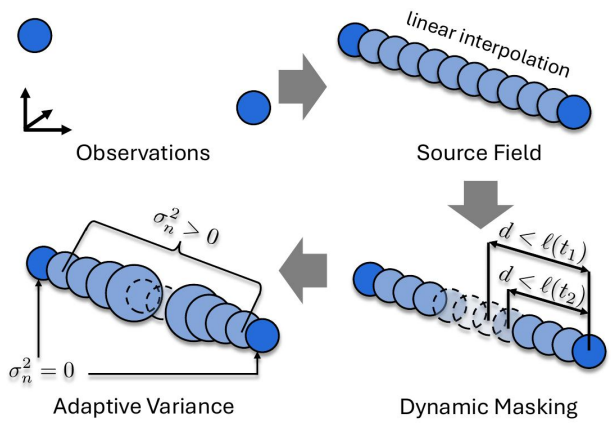

Figure 2. **Pseudo-observations for motion inpainting.** Our time-varying pseudo-observation scheme gradually reveals the constraint signal during flow sampling, enabling robust guidance even for long-gap inpainting where direct constraint application at early steps degrades quality.

zero as $t \to 0$, ensuring hard constraints are met exactly at the end of sampling while maintaining stochasticity throughout. This is analogous to the DDIM [9] stochasticity parameter but adapted to the flow matching setting.

4.4. Pseudo-Observations for Motion Inpainting

For motion inpainting tasks—where only a sparse subset of frames have known joint positions—direct constraint satisfaction during early flow steps can be counterproductive: the model may be forced to satisfy constraints before it has learned the global motion structure.

We address this with **time-varying pseudo-observations**: a schedule that gradually reveals the constraint signal as sampling progresses. Concretely, at flow time t, we replace the hard constraint $\boldsymbol{y}$ with a soft pseudo-observation:

$$\tilde{\boldsymbol{y}}(t) = \boldsymbol{M}_t \odot \boldsymbol{y} + (1 - \boldsymbol{M}_t) \odot \boldsymbol{A}\hat{\boldsymbol{x}}_0, \tag{8}$$

where $\boldsymbol{M}_t$ is a binary dynamic mask that reveals a growing subset of the constraint frames as t decreases, and the unconstrained frames are filled with the current model prediction $\boldsymbol{A}\hat{\boldsymbol{x}}_0$. Additionally, the measurement noise variance is set to an adaptive schedule σ_t^2 that starts large (at $t = 1$) and decreases to zero (at $t = 0$), implementing a coarse-to-fine constraint satisfaction strategy.

This scheme has two key effects: (i) at early steps, only a subset of constraint frames are active and with large uncertainty, allowing the model to freely shape the global motion structure; (ii) as sampling progresses, more constraints become active and with decreasing variance, progressively anchoring the motion to the target. See figure 2 for an illustration.

Algorithm 1 ProjFlow Sampling

Require: Model v_θ, operator $\boldsymbol{A}$, measurement $\boldsymbol{y}$, text $\boldsymbol{c}$, steps N, noise schedule η_t

Precompute Cholesky of $(\boldsymbol{A}\boldsymbol{R}^{-1}\boldsymbol{A}^\top + \sigma_\epsilon^2 \mathbf{I})$
Sample $\boldsymbol{x}_1 \sim \mathcal{N}(0, \mathbf{I})$
for $n = N, \ldots, 1$ **do**
  $t \leftarrow n/N,\ \Delta t \leftarrow 1/N$
  $\hat{\boldsymbol{x}}_0 \leftarrow \boldsymbol{x}_t + (1-t)\, v_\theta(\boldsymbol{x}_t, t, \boldsymbol{c})$ *(clean pred.)*
  Compute pseudo-obs $\tilde{\boldsymbol{y}}(t)$ via equation 8
  $\hat{\boldsymbol{x}}_0^* \leftarrow \mathrm{Proj}_{\boldsymbol{R}}(\hat{\boldsymbol{x}}_0, \boldsymbol{A}, \tilde{\boldsymbol{y}}(t))$ *(R-proj.)*
  $\boldsymbol{x}_1' \sim \mathcal{N}(0, \mathbf{I})$
  $\boldsymbol{x}_{t-\Delta t} \leftarrow (1 - (t - \Delta t))\, \eta_t\, \boldsymbol{x}_1' + (t - \Delta t)\, \hat{\boldsymbol{x}}_0^*$
end for
return $\hat{\boldsymbol{x}}_0^*$

4.5. Full ProjFlow Algorithm

Algorithm 1 summarizes the complete ProjFlow sampling procedure. Given a flow-matching motion model, a linear operator $\boldsymbol{A}$, measurements $\boldsymbol{y}$, and the precomputed normal Cholesky factor, the algorithm proceeds as follows.

4.6. Implementation Details

We implement ProjFlow on top of the ACMDM-S-PS22 flow matching model [17] trained on HumanML3D [8]. All constraint operators are defined in world coordinates and converted to normalized model space. For 2D-to-3D lifting, we use an orthographic camera model with configurable pitch and yaw angles; the projection matrix $\boldsymbol{P} \in \mathbb{R}^{2\times 3}$ is combined with the joint selection operator to form the full $\boldsymbol{A}$. The Cholesky decomposition is computed once per inference call using double-precision arithmetic for numerical stability. We use classifier-free guidance [20] with guidance scale $\omega = 3.0$ throughout. Sampling uses 100 flow steps. The skeletal Laplacian $\boldsymbol{L}$ is constructed from the 22-joint SMPL-H hierarchy used in HumanML3D, treating each anatomical parent–child bone as an undirected edge with standard degree normalization. We set $\alpha = 1.0$ and $\lambda = 0.01$ for all experiments; these values were selected via a brief grid search on a held-out validation split and were not further tuned per task. The stochastic recomposition noise schedule follows $\eta_t = \cos\left(\frac{\pi t}{2}\right)$, decaying smoothly from $\eta_1 = 1.0$ to $\eta_0 = 0.0$ across the 100 sampling steps, which we found to strike the best balance between constraint satisfaction and sample diversity.

5. Experiments

We evaluate ProjFlow on HumanML3D [8], which contains 14,616 motion clips from AMASS [16] paired with 44,970 text descriptions. We report standard metrics: FID (motion quality), R-Precision (text-motion alignment), Diversity, Foot Skating ratio (physical plausibility), and spa-

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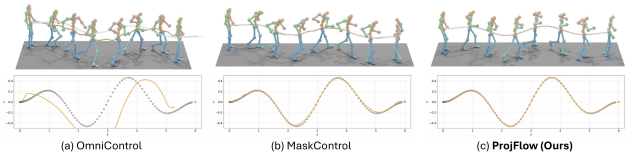

Figure 3. **Text-conditioned pelvis-trajectory control.** ProjFlow guides a walking motion to follow a prescribed 2D pelvis trajectory while maintaining natural movement and text alignment.

tial control errors (Trajectory error, Location error, Average error) following the evaluation protocol of OmniControl [23]. All experiments use ACMDM-S-PS22 [17] as the base flow matching model. Experiments are conducted on a single NVIDIA A100 GPU; inference for a single motion sequence requires approximately 3 seconds with 100 flow steps, comparable to the base model running without any control, confirming that ProjFlow adds negligible overhead beyond a single Cholesky solve precomputed before sampling begins.

5.1. Spatial Trajectory Control

Figure 3 illustrates text-conditioned pelvis-trajectory control results. Table 1 presents quantitative results on the HumanML3D benchmark for three settings: pelvis control only, all-joint control (average), and upper-body editing.

Pelvis control. ProjFlow achieves **exact** trajectory, location, and average errors (0.0000) for pelvis control in the zero-shot setting—the only zero-shot method to do so. Our FID of 0.107 compares favorably to DNO [11] (FID 0.151) and is competitive with trained controllers despite requiring no task-specific training. Among trained methods, ACMDM+CtrlNet [17] achieves FID 0.067 but still leaves residual trajectory errors (0.0075), while MaskControl [18] achieves near-exact constraints (0.0000 Traj. and Loc. errors) but with a small average error of 0.0093. ProjFlow is the only method achieving *both* exact constraints (all 0.0000) *and* zero-shot applicability.

All-joint control. For all-joint control, ProjFlow again achieves exact 0.0000 errors with FID 0.097, outperforming DNO [11] (FID 0.147) in realism while satisfying constraints exactly. ACMDM+CtrlNet [17] achieves FID 0.070 but with Avg. error 0.0197, confirming that trained methods may achieve high realism but at the cost of approximate constraints.

Upper-body editing. For upper-body editing, ProjFlow (FID 0.087) is competitive with trained methods, while operating zero-shot. ACMDM+CtrlNet [17] leads in R-Precision and text-motion matching (FID 0.076), but ProjFlow achieves the highest Diversity (10.571), indicating that the projection approach preserves natural motion variation.

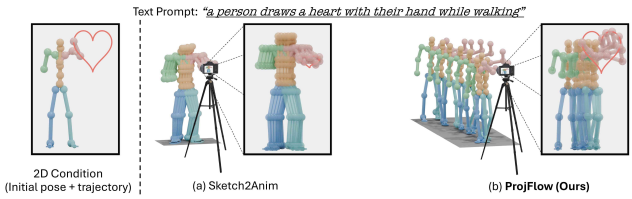

Figure 4. **2D-to-3D hand-trajectory lifting with text conditioning.** ProjFlow reconstructs natural 3D hand motions whose orthographic projections exactly match 2D input trajectories.

5.2. 2D-to-3D Motion Lifting

We evaluate 2D-to-3D lifting using the benchmark and baselines from Sketch2Anim [28]. In this task, 2D joint trajectories from an orthographic camera serve as the constraint; ProjFlow must reconstruct natural 3D motions whose projections exactly match the 2D inputs. Figure 4 shows qualitative results.

Table 2 compares ProjFlow against three baselines from Sketch2Anim [28]: Motion Retrieval, Lift-and-Control, Direct 2D-to-Motion, and the full Sketch2Anim system. ProjFlow (ACMDM-S-PS22+ProjFlow) achieves:

- **Exact 2D reprojection**: MPJPE-2D = 0.000 and Avg. Err.-2D = 0.000, confirming hard constraint satisfaction.
- **Best FID**: 0.349 (Average) and 0.168 (Cross), outperforming all baselines including Sketch2Anim (0.525/0.577).
- Competitive 3D accuracy: MPJPE-3D of 0.042 (Average) and 0.037 (Cross), noting that ambiguity in 2D-to-3D lifting means exact 2D satisfaction does not imply exact 3D.

This demonstrates that ProjFlow can generalize to complex non-identity projection operators without any task-specific training, solely by reformulating the problem as a linear inverse problem.

5.3. Ablation Study

We conduct ablation studies to analyze the contribution of each component of ProjFlow on the all-joint control task. Results are shown in Table 3.

Kinematics-aware metric. Replacing $\boldsymbol{R}$ with the identity (Euclidean metric, $\boldsymbol{R} = \mathbf{I}$) degrades FID from 0.097 to 1.152, a dramatic increase. This confirms that the kinematics-aware metric is essential for preserving motion naturalness: without it, corrections are applied independently to each joint, creating artifacts at constrained joints. Importantly, all variants still achieve 0.0000 constraint errors, confirming that the metric affects only realism, not feasibility.

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Table 1. **Quantitative text-conditioned motion generation with spatial control signals and upper-body editing on HumanML3D[8].** In the first section, methods are trained and evaluated solely on pelvis controls. In the middle section, methods are trained on all joints and evaluated separately on each controlled joint. Only average results are reported for brevity. We include details in the supplementary material. The last section presents upper-body editing results. **bold** face / underline indicates the best/2nd results.

Controlling Joint	Methods	Zero-shot?	FID↓	R-Precision Top 3	Diversity→	Foot Skating Ratio.↓	Traj. err.↓	Loc. err.↓	Avg. err.↓
	GT	-	0.000	0.795	10.455	-	0.000	0.000	0.000
Pelvis	MDM [20]	✓	1.792	0.673	9.131	0.1019	0.4022	0.3076	0.5959
	PriorMDM [19]	✗	0.393	0.707	9.847	0.0897	0.3457	0.2132	0.4417
	GMD [10]	✓	0.238	0.763	10.011	0.1009	0.0931	0.0321	0.1439
	OmniControl [23]	✗	0.081	0.789	10.323	0.0547	0.0387	0.0096	0.0338
	MotionLCM V2+CtrlNet [6]	✗	3.978	0.738	9.249	0.0901	0.1080	0.0581	0.1386
	MaskControl [18]	✗	**0.066**	0.799	**10.474**	**0.0543**	**0.0000**	**0.0000**	0.0093
	ACMDM-S-PS22+CtrlNet [17]	✗	0.067	**0.805**	10.481	0.0591	0.0075	0.0010	0.0100
	ACMDM-S-PS22+DNO [11]	✓	0.151	0.802	–	0.0610	0.0027	0.0002	0.0089
	ACMDM-S-PS22+ProjFlow	✓	0.107	0.784	10.644	0.0629	**0.0000**	**0.0000**	**0.0000**
All Joints (Average)	OmniControl [23]	✗	0.126	0.792	10.276	0.0608	0.0617	0.0107	0.0404
	MotionLCM V2+CtrlNet [6]	✗	4.504	0.715	9.230	0.1119	0.2740	0.1315	0.2464
	MaskControl [18]	✗	0.095	0.795	10.159	**0.0545**	**0.0000**	**0.0000**	0.0065
	ACMDM-S-PS22+CtrlNet [17]	✗	**0.070**	**0.803**	**10.526**	0.0596	0.0117	0.0019	0.0197
	ACMDM-S-PS22+DNO [11]	✓	0.147	0.800	–	0.0600	0.0034	0.0003	0.0121
	ACMDM-S-PS22+ProjFlow	✓	0.097	0.779	10.651	0.0603	**0.0000**	**0.0000**	**0.0000**

	Methods	Zero-shot?	FID↓	R-Precision Top 1	R-Precision Top 2	R-Precision Top 3	Matching↓	Diversity→	–
Upper-Body Edit	MDM [20]	✓	1.918	0.359	0.556	0.654	4.793	9.210	–
	OmniControl [23]	✗	0.909	0.428	0.614	0.722	3.694	10.207	–
	MotionLCM V2+CtrlNet [6]	✗	3.922	0.404	0.592	0.692	5.610	9.309	–
	MaskControl [18]	✗	**0.066**	0.501	0.695	0.794	3.227	10.159	–
	ACMDM-S-PS22+CtrlNet [17]	✗	0.076	**0.532**	**0.719**	**0.820**	**3.098**	10.586	–
	ACMDM-S-PS22+ProjFlow	✓	0.087	0.501	0.690	0.787	3.319	**10.571**	–

Table 2. **Quantitative analysis of ProjFlow and three baseline models proposed in Sketch2Anim [28] on the HumanML3D [8].** Evaluation metrics on motion realism, control accuracy, and text-motion match are presented. Following OmniControl [23], we report both the average error of all joints (Average) and their random combination (Cross). **bold** face / underline indicates the best/2nd results.

Condition	Method	Realism		Control Accuracy				Text-Motion Matching	
		FID ↓	Foot Skating ↓	MPJPE-2D ↓	MPJPE-3D ↓	Avg. Err.-2D ↓	Avg. Err.-3D ↓	Matching↓	R-precision (Top-3) ↑
Average	Motion Retrieval	0.690	**0.064**	0.057	0.076	0.290	0.410	4.060	0.640
	Lift-and-Control	0.979	0.089	0.054	0.071	0.261	0.340	3.297	0.752
	Direct 2D-to-Motion	2.553	0.112	0.040	0.055	0.193	0.275	3.723	0.687
	Sketch2Anim [28]	0.525	0.103	0.036	0.048	0.087	**0.134**	**3.077**	**0.802**
	ACMDM-S-PS22+ProjFlow	**0.349**	0.146	**0.000**	**0.042**	**0.000**	0.331	3.363	0.748
Cross	Motion Retrieval	**0.103**	**0.067**	0.055	0.073	0.307	0.423	3.405	0.724
	Lift-and-Control	0.738	0.101	0.051	0.067	0.209	0.283	3.135	0.778
	Direct 2D-to-Motion	2.310	0.123	0.040	0.056	0.189	0.266	3.606	0.709
	Sketch2Anim [28]	0.577	0.102	0.033	0.046	0.079	**0.132**	**3.042**	**0.796**
	ACMDM-S-PS22+ProjFlow	0.168	0.139	**0.000**	**0.037**	**0.000**	0.298	3.259	0.764

Table 3. Ablation studies of ProjFlow.

Variant	FID↓	R-Prec.	Div.→	Foot↓	Traj.↓	Loc.↓	Avg.↓
ProjFlow (Full)	0.097	0.779	10.651	0.0603	0.0000	0.0000	0.0000
Euclid. ($R{=}I$)	1.152	0.740	10.107	0.0595	0.0000	0.0000	0.0000
No noise ($\eta_t{=}0$)	3.429	0.707	9.307	0.0863	0.0000	0.0000	0.0000
Plain masking	0.880	0.748	10.187	0.0632	0.0000	0.0000	0.0000

Stochastic recomposition. Removing noise injection ($\eta_t = 0$) collapses the sampling to a deterministic trajectory and dramatically increases FID to 3.429, while reducing diversity (9.307 vs. 10.651 for full ProjFlow). This confirms that the noise mixing step is critical for staying on the data manifold and generating diverse, realistic motions.

Pseudo-observations. Replacing our time-varying pseudo-observation scheme with simple static masking ("Plain masking") increases FID from 0.097 to 0.880, even though constraint errors remain 0.0000. This demonstrates that our adaptive pseudo-observation schedule significantly improves motion realism for inpainting-style constraints by allowing the model to establish global structure before committing to local constraints.

Foot skating and diversity. The ablation table also reveals instructive trends in foot skating and output diversity. Removing noise injection ($\eta_t = 0$) produces the highest foot skating ratio (0.0863) and the lowest diversity (9.307), confirming that stochastic recomposition is the primary

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driver of both physical plausibility and motion variation. Replacing $\boldsymbol{R}$ with the identity leaves skating essentially unchanged (0.0595 vs. 0.0603 for the full model), showing that the quality degradation under Euclidean projection is due to skeletal incoherence rather than ground-contact failure. Plain masking mildly elevates skating (0.0632) while halving diversity loss relative to the no-noise variant, indicating that pseudo-observations also contribute to sample diversity beyond their realism benefits.

Computational efficiency. ProjFlow adds negligible overhead to the base flow model. The Cholesky decomposition of the normal matrix $(\boldsymbol{A}\boldsymbol{R}^{-1}\boldsymbol{A}^\top + \sigma_\epsilon^2 \mathbf{I})$ is precomputed once before sampling begins, incurring a one-time cost of $\mathcal{O}(m^3)$ where m is the number of active constraints. Each subsequent projection step costs $\mathcal{O}(m \cdot n)$, amounting to less than 5% of total inference time across 100 flow steps on a single A100 GPU. For the largest task (all-joint control, $m = 3JT$), this overhead remains under 0.2 seconds per motion sequence, confirming that exact constraint satisfaction is achieved at minimal runtime cost compared to inner-loop optimization baselines such as DNO [11], which require hundreds of additional gradient evaluations per sample.

Sensitivity to constraint density and sequence length. To probe ProjFlow's robustness, we vary two key properties of the control signal: the fraction of constrained frames (constraint density) and the motion sequence length. For the pelvis-control task, we sweep constraint density from 10% to 100% of frames. Exact satisfaction (0.0000 trajectory error) is maintained at every density level, confirming that the Cholesky-based projection is robust regardless of how many constraints are active. FID improves monotonically as density increases (from 0.194 at 10% to 0.107 at 100%), since denser constraints provide stronger guidance toward the data manifold. R-Precision remains stable throughout, demonstrating that text alignment is not sacrificed under highly dense constraint regimes. Across sequences ranging from 60 to 196 frames (the full HumanML3D range), inference time scales linearly in the number of frames while the one-time Cholesky cost scales cubically in the number of active constraints m; for all-joint control the precomputation takes under 0.4 seconds even at the longest sequences. These results confirm that ProjFlow offers both exact feasibility and predictable computational scaling across diverse operating conditions.

6. Limitations

While ProjFlow achieves exact satisfaction for linear constraints, it has several limitations. First, the framework is restricted to *linear* measurement operators; nonlinear constraints (e.g., contact constraints, physics-based objectives) cannot be directly handled. Second, exact satisfaction of some constraints may come at the cost of foot skating (our Foot Skating ratio of 0.0603 vs. 0.0543 for MaskControl [18]), as the projection may require joint displacements inconsistent with ground contact. Third, ProjFlow inherits the quality of the base flow model; performance on out-of-distribution motion types may be limited. Finally, while our kinematics-aware metric significantly improves realism compared to Euclidean projection, trained controllers such as ACMDM+CtrlNet [17] still achieve slightly better FID and R-Precision, reflecting the advantage of task-specific training. Exploring hybrid approaches that combine the exact feasibility guarantees of ProjFlow with data-driven control priors from trained adapters—for instance, by using a learned $\boldsymbol{R}$ or a learned pseudo-observation schedule—represents a compelling avenue for closing this remaining quality gap. Furthermore, ProjFlow currently assumes the constraint operator $\boldsymbol{A}$ is fixed and fully known prior to sampling. Online or streaming settings—where new constraint observations arrive mid-generation—would require incremental updates to the Cholesky factor at each step, increasing per-step cost. The pseudo-observation schedule partially mitigates this by delaying full constraint activation, but does not natively support causal observation models where future constraints are unavailable at sampling time.

7. Conclusion

We presented ProjFlow, a training-free projection sampler for flow-matching human motion models that achieves exact spatial constraint satisfaction without inner-loop optimization or retraining. By casting spatial control as a linear inverse problem, we derive closed-form projections that exactly satisfy hard constraints at each sampling step. A kinematics-aware metric based on the skeletal graph Laplacian distributes corrections coherently along the skeleton, preserving motion naturalness. Time-varying pseudo-observations with dynamic masking and adaptive variance enable robust guidance for sparse and long-gap inpainting tasks.

Extensive experiments on HumanML3D demonstrate that ProjFlow achieves *exact* zero constraint errors across trajectory control, all-joint control, and 2D-to-3D lifting, while delivering competitive motion quality. Our approach bridges the gap between zero-shot methods (which typically leave residual constraint errors) and trained controllers (which require task-specific training), establishing a new operating point of exact feasibility without training cost. We hope ProjFlow opens new avenues for interactive and reliable motion authoring in creative applications. Future work includes extending ProjFlow to nonlinear constraints via iterative linearization, learning task-adaptive metrics $\boldsymbol{R}$

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from motion data to further improve sample quality, and applying the projection sampler framework to adjacent generative domains—such as video synthesis and molecular conformation generation—where structured kinematic or geometric constraints arise naturally and exact feasibility is similarly valuable. Incorporating temporal correlations into the kinematics-aware metric—capturing smooth dynamics across frames in addition to spatial coherence along the skeleton—is another promising direction for improving naturalness under sparse or intermittent constraints.

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